Session 4: Sequence Alignment

KB8029 — Bioinformatics & Deep Learning

Arne Elofsson

SciLifeLab

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Log in to the lecture quiz

Log in to the quiz

Scan the code, or go to quiz.bioinfo.se/live

Log in with your university account

Answering counts as being at the lecture: up to 3 bonus points on the exam

Right or wrong is never graded, only whether you answered. Results are shown without names.

quiz.bioinfo.se/live

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Warm-up: what do you know already?

3 quick questions

On the topics of today, one at a time, about 30 seconds each

Answer on your own, and guess if you do not know

It shows me where to start; right or wrong is never graded

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Two genes in different species descend from one gene in their last common ancestor, split by speciation. They are ...

A paralogs

B orthologs

C analogs

D unrelated

Answer now
Scan the code, or go to
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An affine gap penalty has g_open = −10 and g_ext = −1. What does one gap of length 3 cost?

A −12

B −13

C −30

D −3

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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A column of a multiple alignment has the same amino acid in every sequence. What is its entropy?

A 0

B 1

C $\log_2 20$, the largest possible

D It depends on the amino acid

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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What the room knows already

Where we start

Which topics most of you know, and which we take slowly today.

Show the results ▸ quiz.bioinfo.se/live/report

Your questions before today

What you asked

The questions you sent in on Canvas before the lecture

The most common ones now; the others in the block where they belong

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In up to three words: what do you want to learn today?

Pairwise alignment

dynamic programming, global and local

Scoring

gaps, BLOSUM, significance

Many sequences

MSA, guide trees, logos

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

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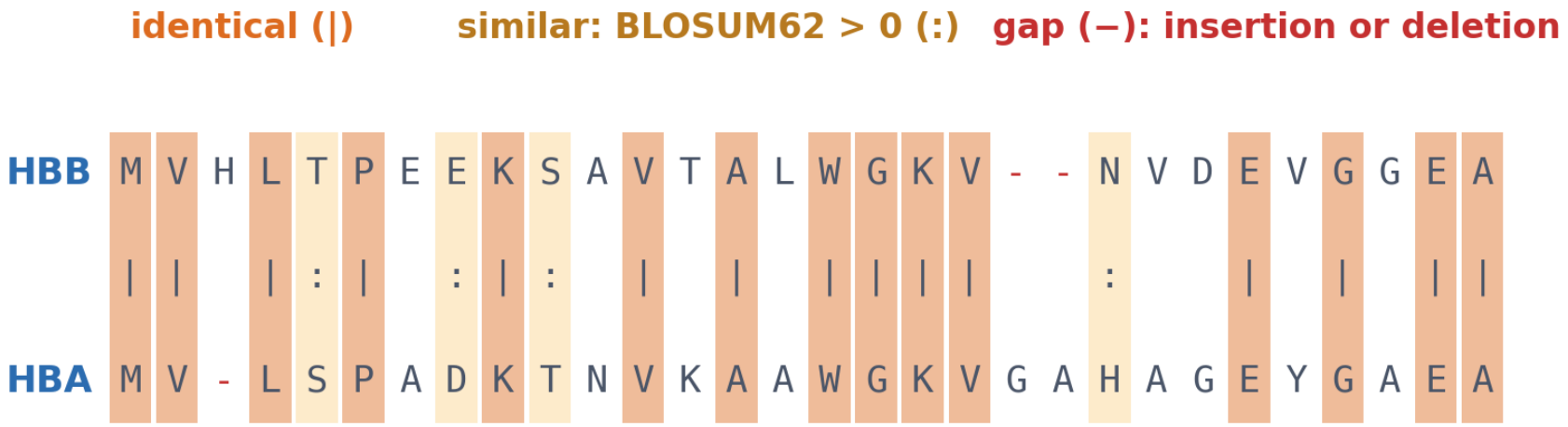
What you want to learn today

What the room wants to learn

Show the results ▸ quiz.bioinfo.se/live/report

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An alignment is a hypothesis: which residues correspond?

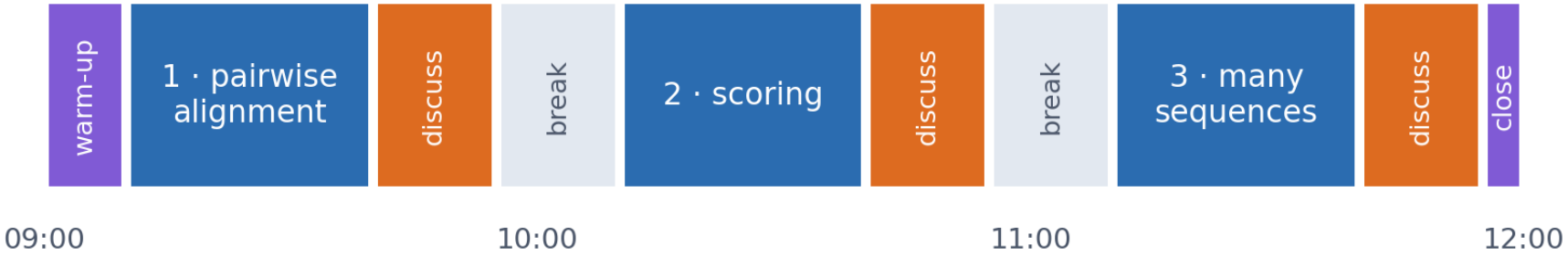


The start of the BLOSUM62 alignment of human β- and α-globin (HBB, HBA). Aligned pairs are substitutions or identities; gaps are insertions or deletions. The score says how plausible this history is under the chosen scoring scheme.

Alignment computed as in the course notebook (BLOSUM62, gap open −10, extend −0.5).

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Three blocks: about 30 min talk and 15 min discussion



Block 1: pairwise alignment by dynamic programming, global and local. Block 2: scoring it, with gap penalties, substitution matrices and significance. Block 3: many sequences, with multiple alignment, trees and logos.

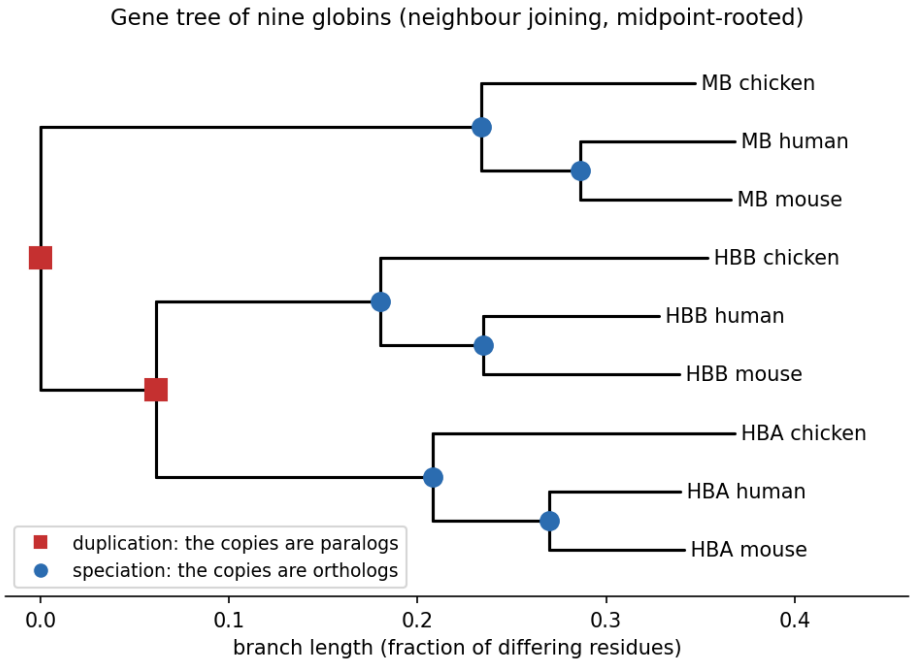
Times are approximate: a 10 min start (warm-up quiz and your questions), two 15 min breaks and a 5 min close. Each block ends with a quiz question: guess, run the notebook and work out why it behaves as it does, discuss, answer again.

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BLOCK 1 · ABOUT 30 MIN

Pairwise alignment needs an algorithm

Homologs: orthologs and paralogs



- **Homology: a common ancestor (yes or no)**
- Identity, similarity: evidence for it
- **Ortholog: split by a speciation**
- **Paralog: split by a duplication**

Gene tree computed in the course notebook; see course.bioinfo.se/session04.html, 'Homology'.

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Where you place the sequences changes the answer

- **Real sequences differ in length**
- Insertions and deletions have accumulated
- Placement is the real problem

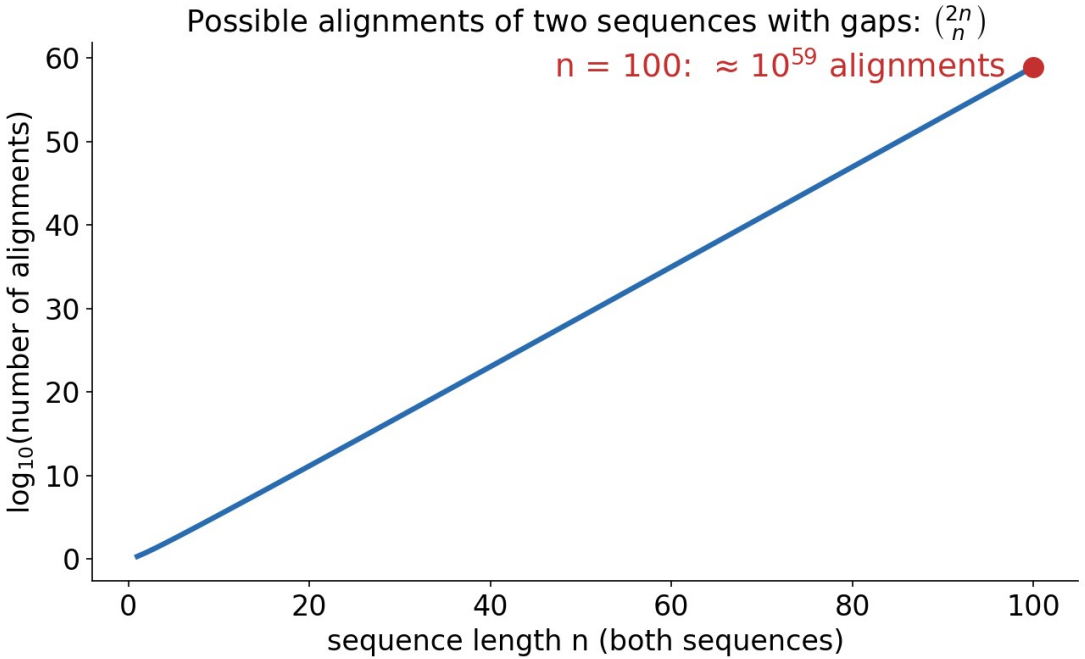
21.5%
identity, aligned correctly

4%
identity, the same two globins placed naively

See course.bioinfo.se/session04.html, 'Why alignment needs an algorithm'.

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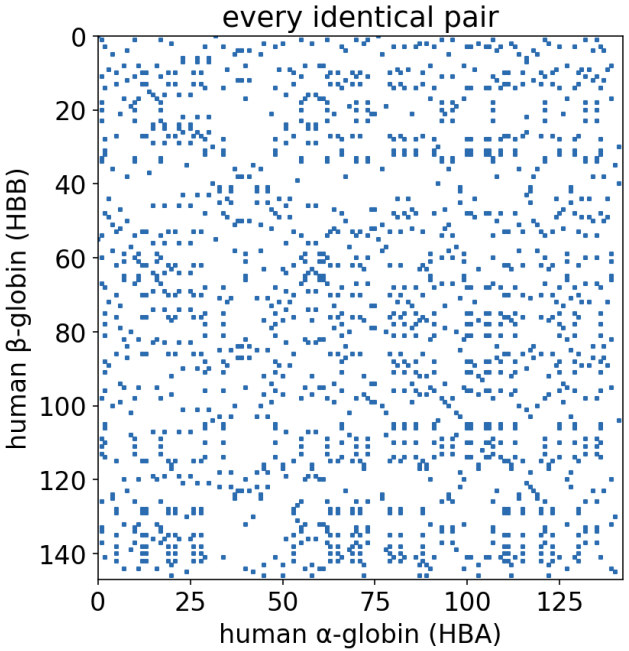
Brute force: try every alignment?

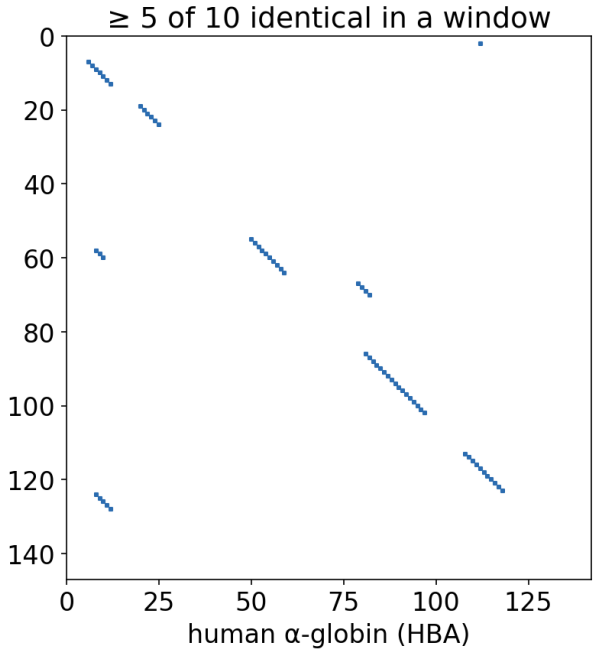


- **n = 100: about 10^{59} alignments**
- Far too many to search
- Dynamic programming: optimal, without checking them all

See course.bioinfo.se/session04.html, 'Why alignment needs an algorithm'.

A dot plot: is there a match, and roughly where?





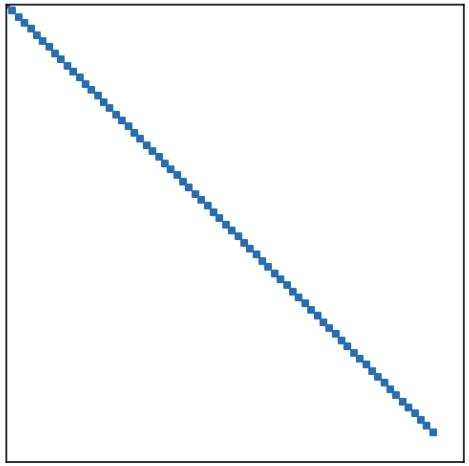
- **A run of matches = a diagonal**
- A window filter removes the noise
- No score, no alignment

HBB and HBA (human) as in the course notebook; see course.bioinfo.se/session04.html, 'Visualising alignments: dot plots'.

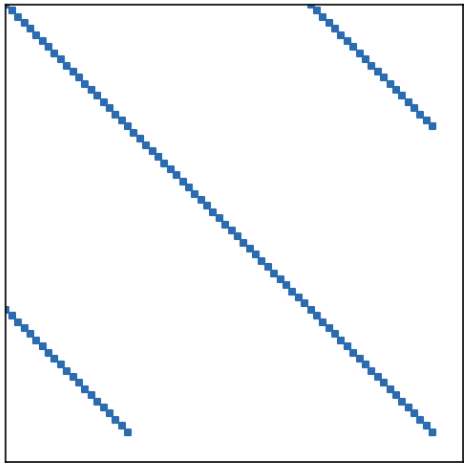
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Reading a dot plot: diagonals, repeats, inversions

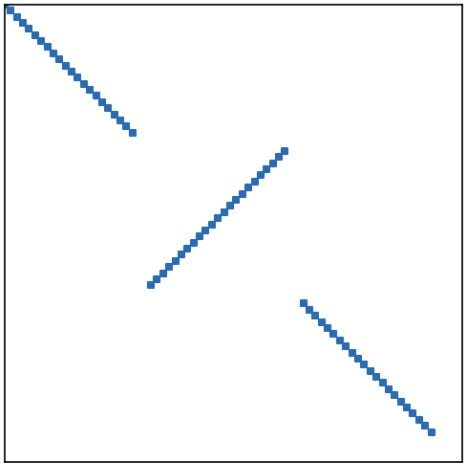
one alignment: a diagonal



a repeat: parallel diagonals



an inversion: a reversed diagonal

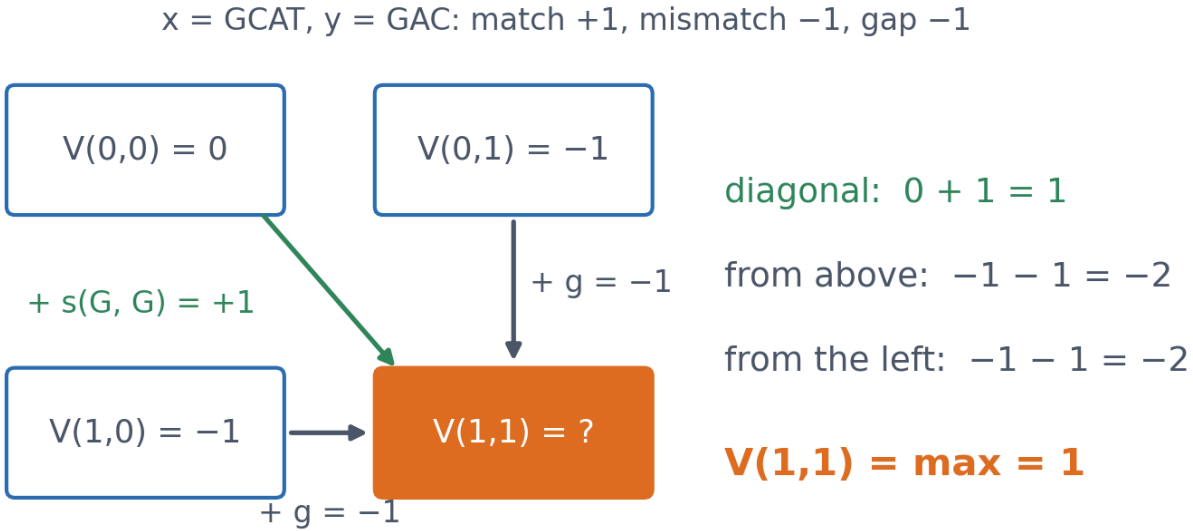


Sequences built from three 25-residue pieces of HBB (window 5, ≥ 4 identical)

One long alignment is one diagonal; a repeated piece adds parallel diagonals; an inverted piece runs the other way. Then a score and an alignment need dynamic programming.

Constructed from pieces of HBB; see course.bioinfo.se/session04.html, 'Visualising alignments: dot plots'.

Each cell: three candidates, keep the largest



diagonal: 0 + 1 = 1

from above: −1 − 1 = −2

from the left: −1 − 1 = −2

V(1,1) = max = 1

- **Diagonal: match or mismatch**
- **Up or left: a gap**
- V(i,j): best score of the two prefixes

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Fill the whole matrix, row by row

	-	G	A	C
-	0	-1	-2	-3
G	-1	1	0	-1
C	-2	0	0	1
A	-3	-1	1	0
T	-4	-2	0	0

- **x = GCAT, y = GAC**
- match +1, mismatch −1, gap −1
- First row and column: −1, −2, −3 …
- **Bottom right: the best score, 0**

See course.bioinfo.se/session04.html, Table 4.1.

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Traceback: follow the choices back

	-	G	A	C
-	**0**	-1	-2	-3
G	-1	**1**	0	-1
C	-2	**0**	0	1
A	-3	-1	**1**	0
T	-4	-2	0	**0**

G C A T

G – A C

$1 - 1 + 1 - 1 = 0$

start bottom right,
follow the choices back

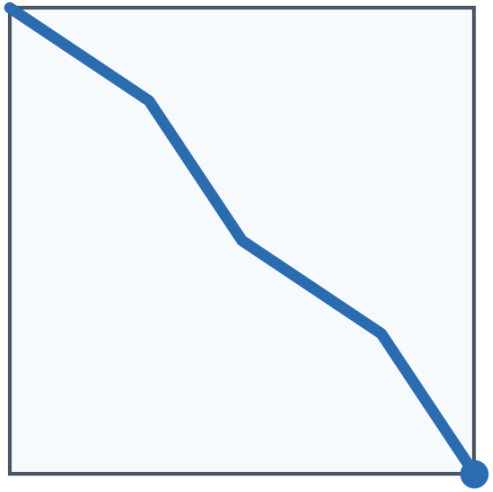
- **G = G, C deleted, A = A, T/C mismatch**
- Score 0, from the corner
- Optimal, guaranteed

See course.bioinfo.se/session04.html, Table 4.1 (the path in bold).

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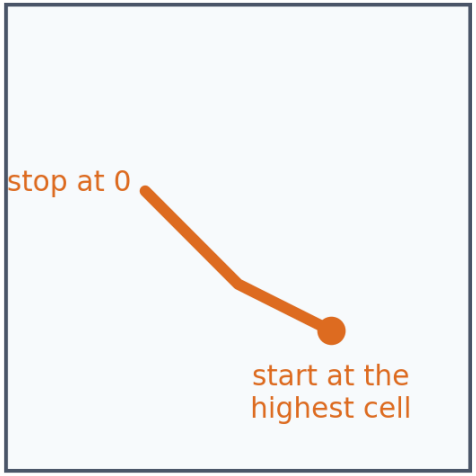
Needleman-Wunsch (global) and Smith-Waterman (local)

global: corner to corner



start the traceback in the corner

local: the best island



- **Global: both sequences, end to end**
- **Local: the best-matching region**
- One change: negative cells → 0

See course.bioinfo.se/session04.html, 'Dynamic programming: Needleman-Wunsch and Smith-Waterman'.

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Which one? A biological choice

Global

two myoglobins

- Expected to correspond along most of their length
- Every residue must be placed

Local

one shared domain, a motif

- Two large multidomain proteins
- A short motif inside a long sequence

See course.bioinfo.se/session04.html, 'Dynamic programming: Needleman-Wunsch and Smith-Waterman'.

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Global or local: which scores higher?

x = ATCG, y = TCG; match +1, mismatch −1, gap −1. Global (Needleman-Wunsch) or local (Smith-Waterman)?

the DP matrix

	-	T	C	G
-				
A				
T				
C				
G				

x = ATCG

y = TCG

match +1
mismatch −1
gap −1

global: every residue
local: may start and end anywhere

A Global 2, local 3

B Both 3: TCG against TCG either way

C Global 3, local 2

D Both 2: the A costs a gap either way

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

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Global or local: which scores higher?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

Global or local: which scores higher?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session04-discussion-1.ipynb

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Global or local: which scores higher?

Answer again

Same question, after the discussion.



quiz.bioinfo.se/live

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Global or local: which scores higher?

Results, and who explains

Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 1 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session04-discussion-1.ipynb

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The answer: global 2, local 3

global: 2

	-	T	C	G
-	**0**	-1	-2	-3
A	**-1**	-1	-2	-3
T	-2	**0**	-1	-2
C	-3	-1	**1**	0
G	-4	-2	0	**2**

```
ATCG
-TCG
```

the A must go against a gap: −1

local: 3

	-	T	C	G
-	0	0	0	0
A	0	0	0	0
T	0	**1**	0	0
C	0	0	**2**	1
G	0	0	1	**3**

```
TCG
TCG
```

the A is left out; negative cells become 0

Global alignment must place every residue, so the leading A goes against a gap (−1). Local alignment sets negative cells to 0 and may start anywhere, so it leaves the A out and scores the pure TCG match. The unfixed notebook had zeros in the first column: a global alignment that could skip the start for free.

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BLOCK 1 DONE

Break

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BLOCK 2 · ABOUT 30 MIN

Scoring: gaps, substitution matrices, significance

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Linear gaps scatter, affine gaps stay together

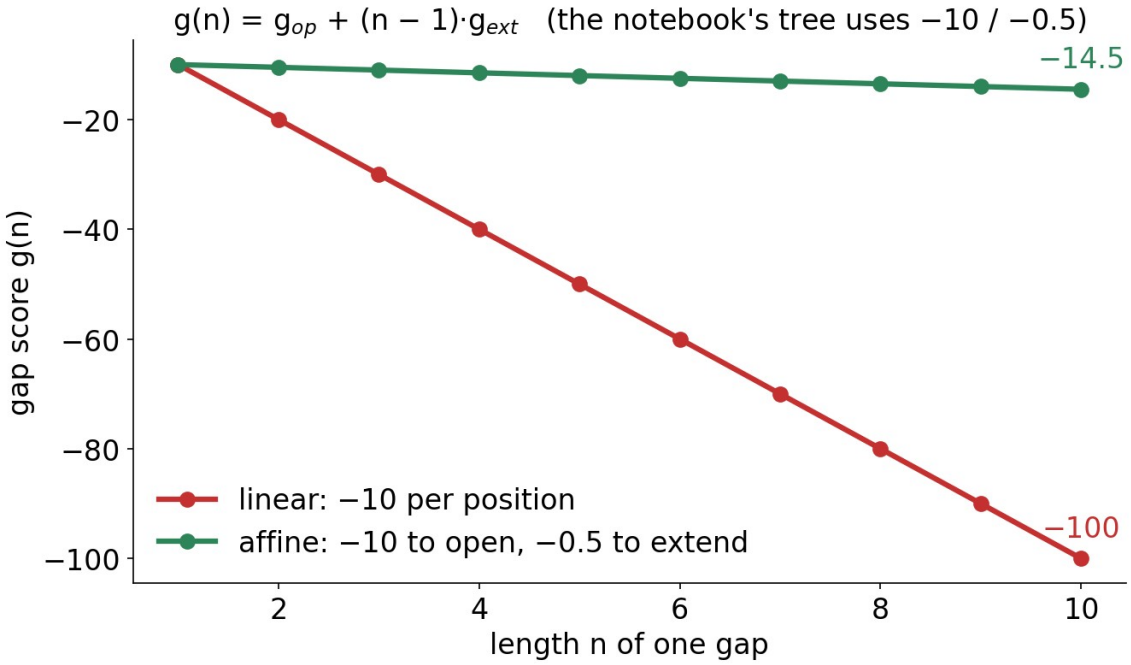


- **Linear: every gap position costs the same**
- Indels often hit whole stretches
- **Affine keeps QQW intact**

See course.bioinfo.se/session04.html, 'Gap penalties: linear vs. affine'.

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Affine: expensive to open, cheap to extend

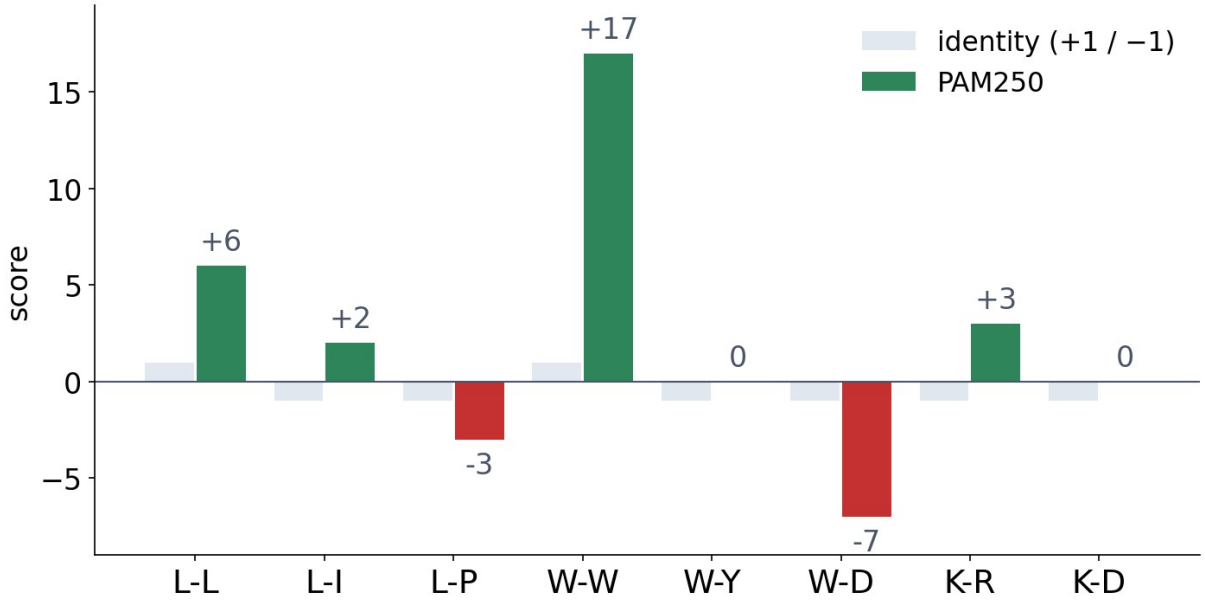


- **g(n) = g_op + (n − 1)·g_ext**
- g_ext ≪ g_op
- One long gap beats many short ones

Gap scores −10 / −0.5 as in the course notebook's pairwise alignments.

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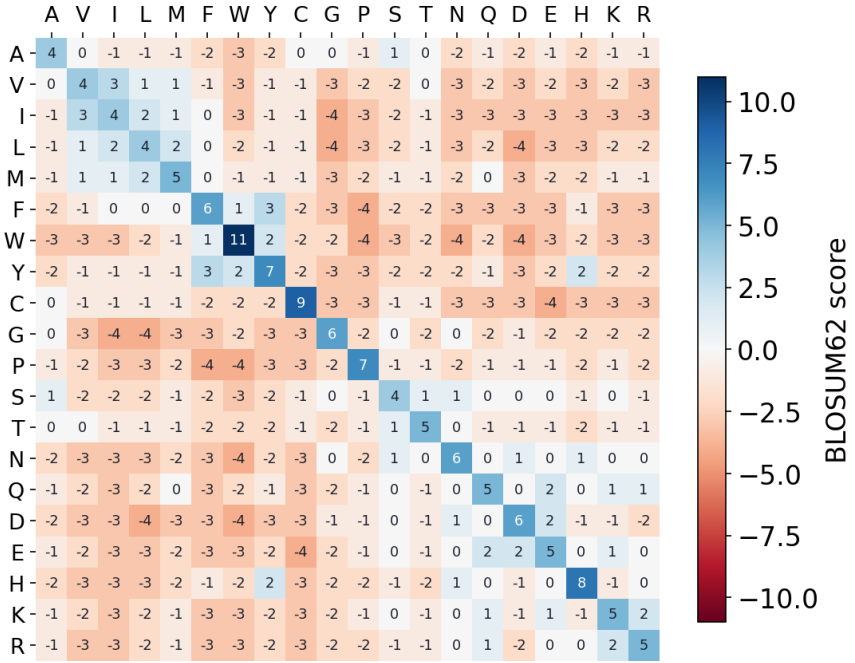
Not every mismatch costs the same



- **Log-odds: seen in relatives vs by chance**
- Positive: more often than chance
- K-R, L-I: conservative, positive

PAM250 scores printed by the course notebook (from Biopython).

BLOSUM62: the whole matrix



- **Diagonal: identities score highest**
- Similar residues: small positive scores
- BLAST's default (Session 5)

BLOSUM62 from Biopython; residues ordered from hydrophobic to charged.

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Two families: PAM and BLOSUM

PAM (Dayhoff 1978)

close relatives, a model

- Global alignments of closely related proteins
- Counts along a phylogenetic tree
- PAM250 extrapolated from short distances

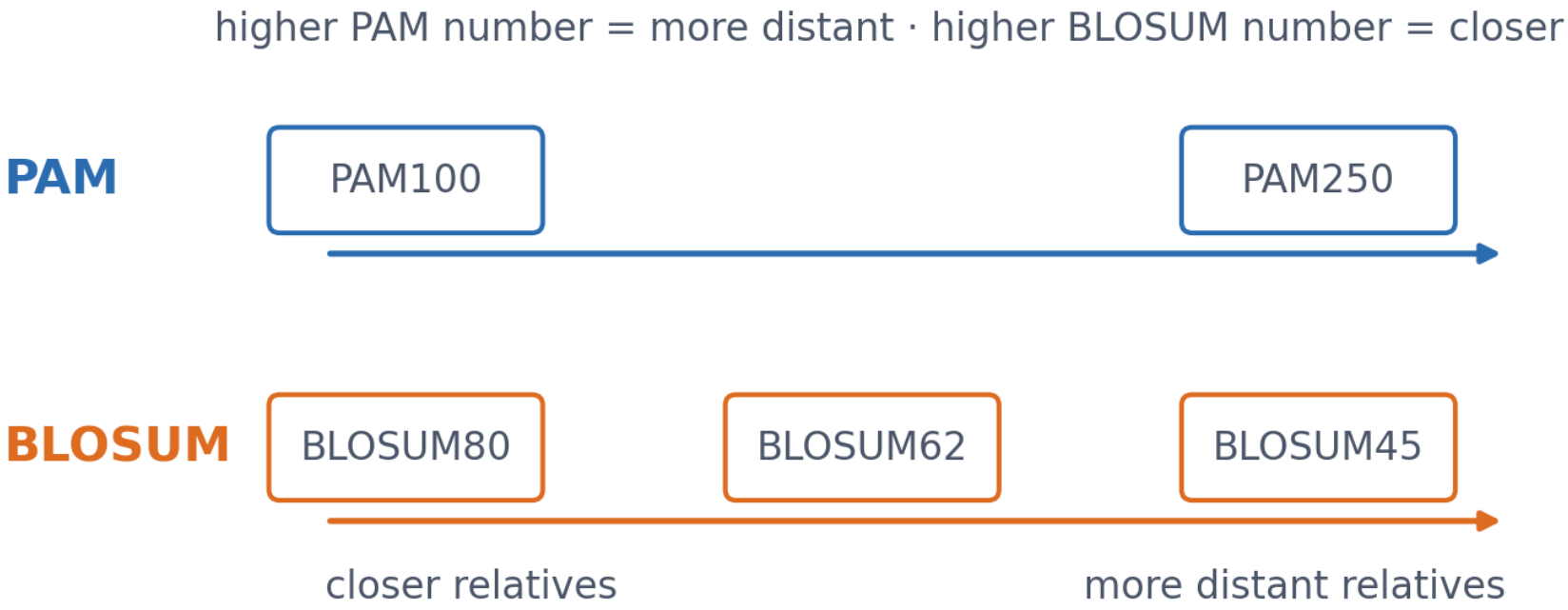
BLOSUM (Henikoff 1992)

distant relatives, counted

- Local, ungapped blocks of distant proteins
- Counted directly: no tree
- Each matrix derived independently

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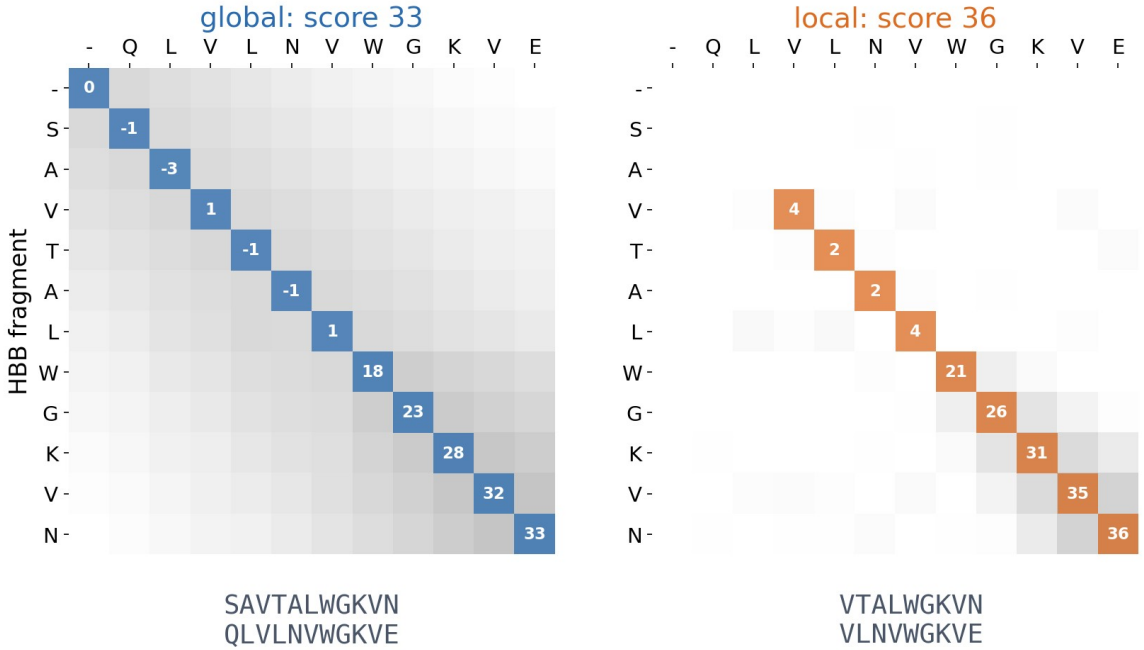
The numbering gotcha



PAM250 suits more divergent sequences than PAM100; BLOSUM80 suits more similar sequences than BLOSUM45. The two scales are not aligned with each other in this picture.

See course.bioinfo.se/session04.html, 'Substitution matrices: PAM and BLOSUM'.

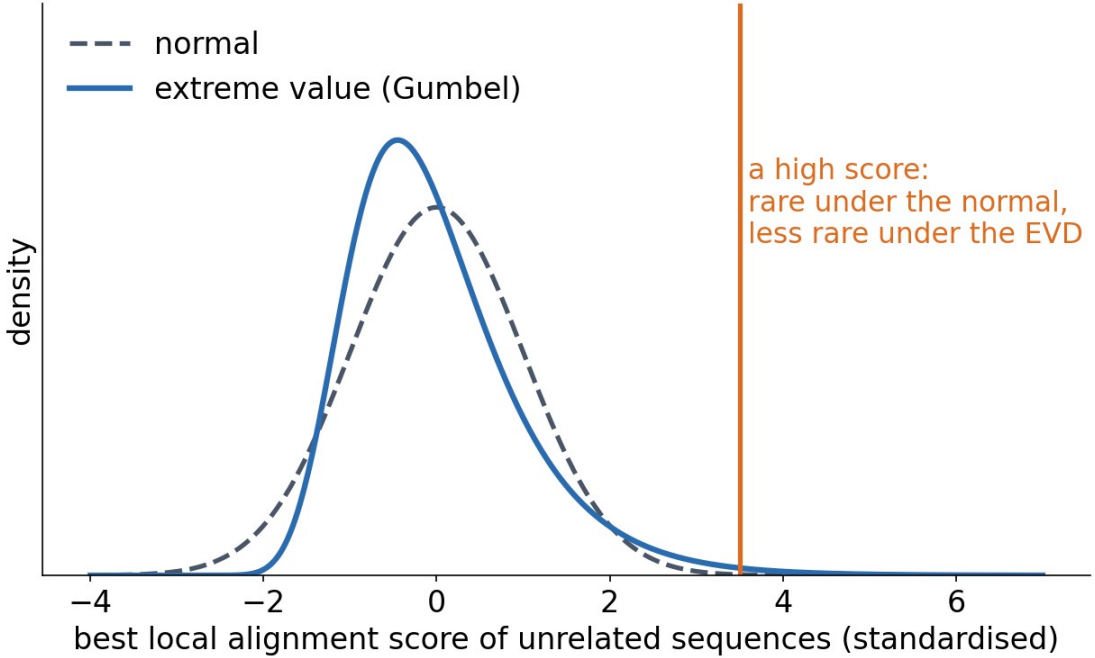
HBB vs myoglobin fragments, scored with PAM250



Global alignment must use all of SAVTALWGKVN and QLVLNVWGKVE (score 33). Local alignment drops the unrelated start and keeps the shared WGKV region (score 36). Gap −8.

Course notebook, 'Global vs. local, side by side'.

Is a high score good, or just chance?



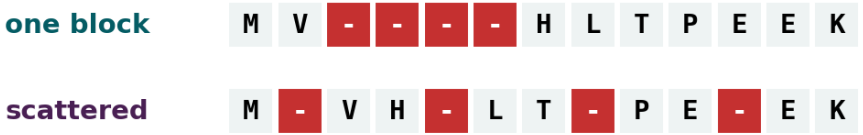
- **No universal threshold**
- Compare with shuffled sequences
- Best local scores: extreme-value distribution
- Session 5: E-values

See course.bioinfo.se/session04.html, 'Is the score significant?'.

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What does affine charge for scattered gaps?

Four gap positions, as one block or as four single gaps. Linear (g = −1): −4 either way. Affine (g_open = −8, g_ext = −1)?




```
linear:  g = −1 per gap position
affine:  g_open = −8, g_ext = −1
         g(n) = g_open + (n − 1) g_ext
```

A −11 block, −32 scattered

B −11 for both, like linear

C −32 block, −11 scattered

D −8 block, −32 scattered

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

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What does affine charge for scattered gaps?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

What does affine charge for scattered gaps?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session04-discussion-2.ipynb

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What does affine charge for scattered gaps?

Answer again

Same question, after the discussion.

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What does affine charge for scattered gaps?

Results, and who explains

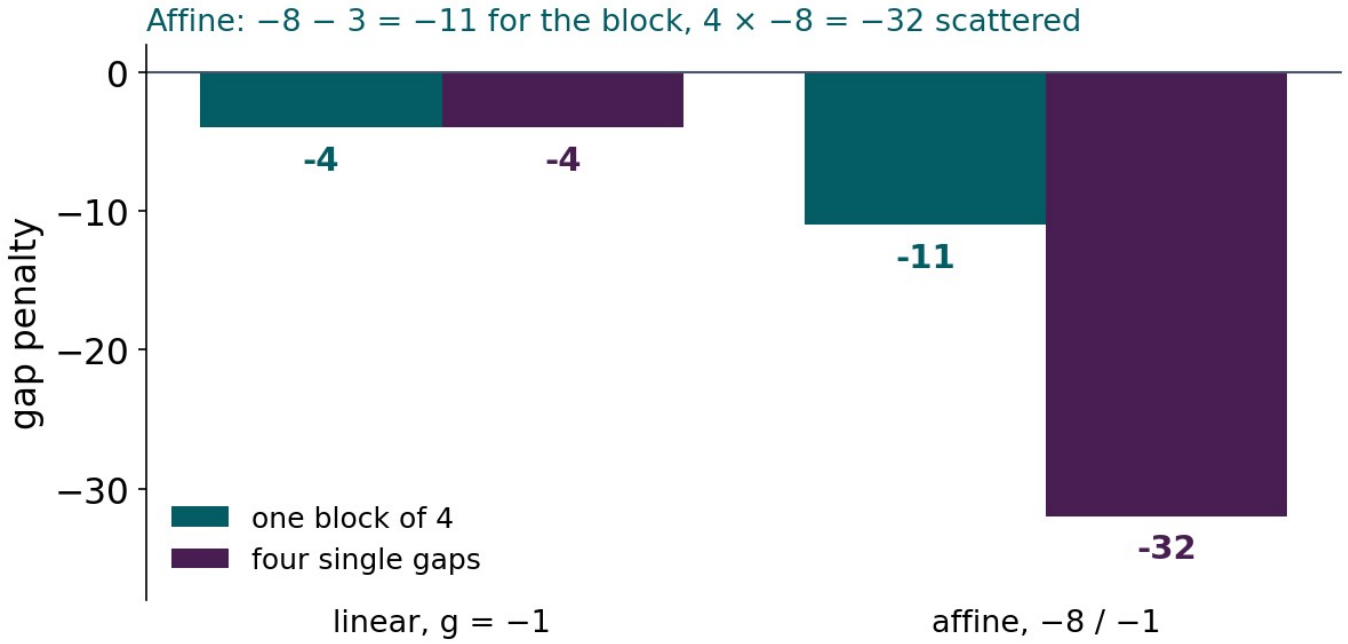
Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 2 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session04-discussion-2.ipynb

The answer: −11 for the block, −32 scattered



Linear charges per gap position, so it cannot tell the two rows apart. Affine charges an opening cost for every separate gap: one block pays it once, four scattered gaps pay it four times. That is why affine penalties give alignments with gaps in blocks.

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BLOCK 2 DONE

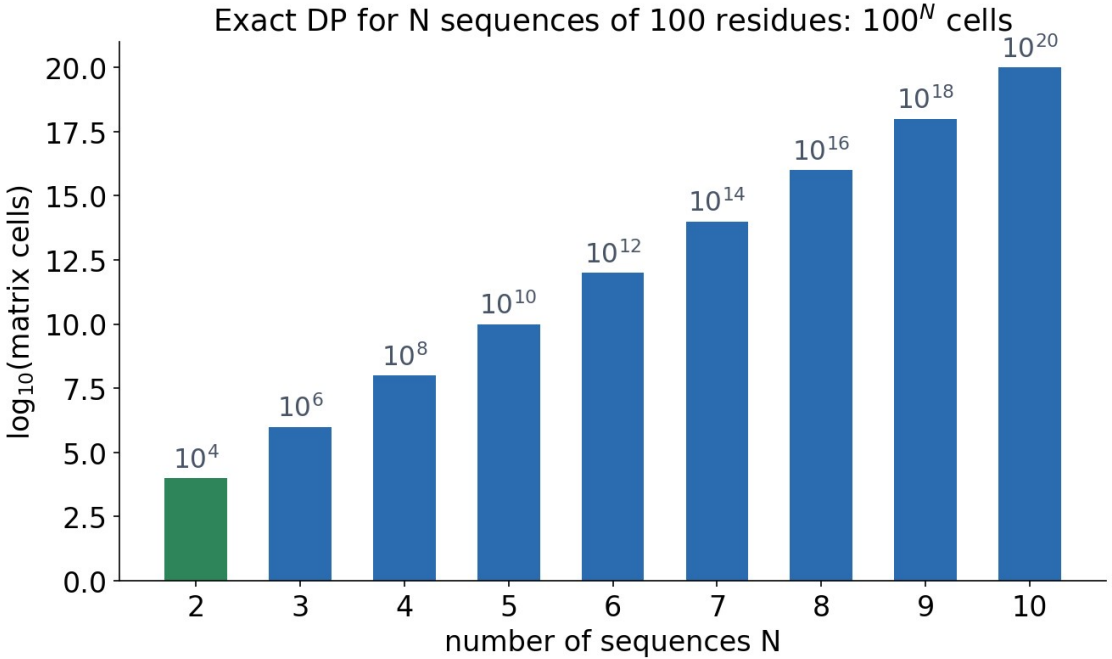
Break

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BLOCK 3 · ABOUT 30 MIN

Many sequences: multiple alignment, trees and logos

Exact DP for many sequences is out of reach



- **The cost grows exponentially with N**
- Fine for a few short sequences
- Practical tools combine three ideas

See course.bioinfo.se/session04.html, 'Multiple sequence alignment'.

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Three ideas behind real MSA tools

Progressive

ClustalW

- Guide tree from pairwise distances
- Closest pair first, then add the rest
- Early mistakes persist

Iterative

MUSCLE

- Start from a progressive alignment
- Re-align parts, keep improvements

Consistency

T-Coffee

- Use agreement among all pairs
- A~B and B~C: prefer A~C

See course.bioinfo.se/session04.html, 'Multiple sequence alignment'.

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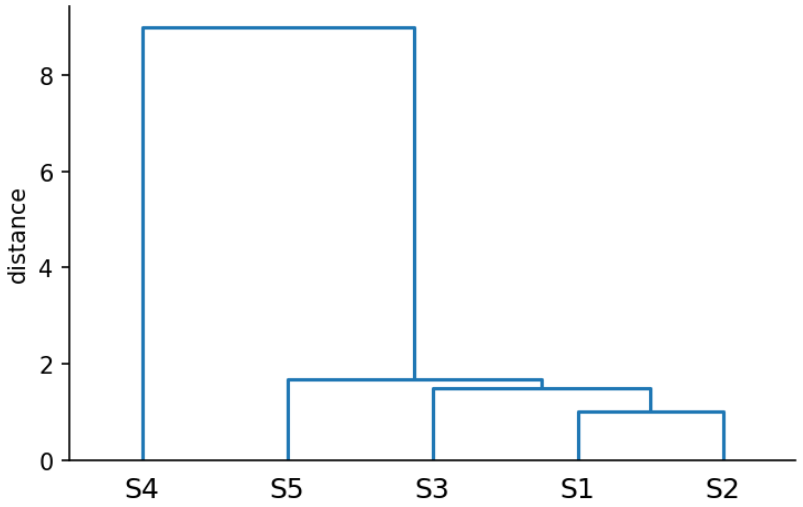
The guide tree decides the order of alignment

Hamming distance, D

	S1	S2	S3	S4	S5
S1	0	1	1	9	1
S2	1	0	2	9	2
S3	1	2	0	9	2
S4	9	9	9	0	9
S5	1	2	2	9	0

```
S1  MKTAVGLTDR
S2  MKTAVGLTDK
S3  MKTAAGLTDR
S4  QWERTYASDF
S5  MKTAVGLADR
```

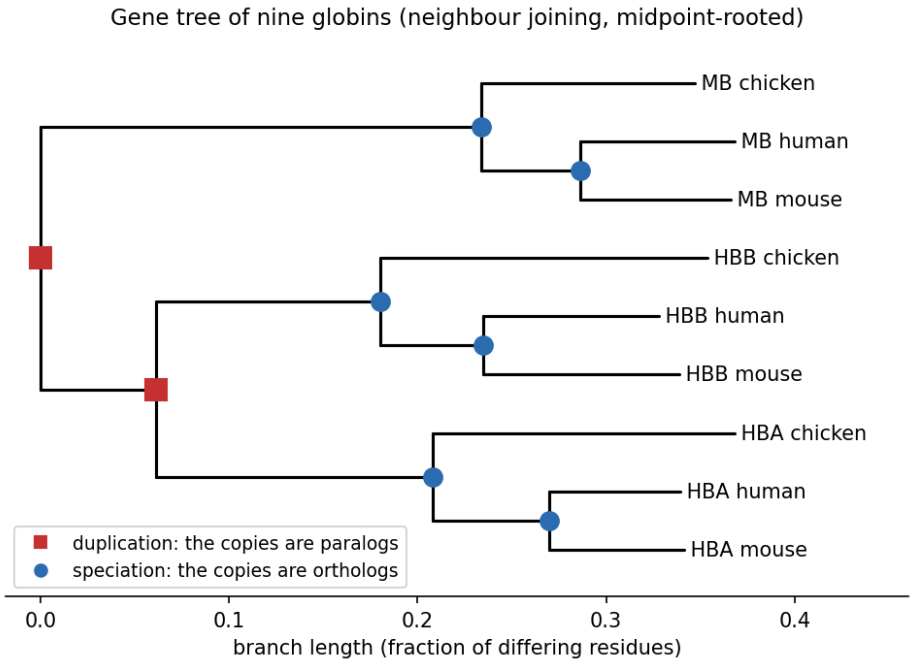
Guide tree (average linkage)



Five toy sequences: every sequence starts as its own cluster, and the closest pair is merged repeatedly. S4 merges last, because it differs clearly from the other four. The merge order is the order of alignment.

See course.bioinfo.se/session04.html, 'Multiple sequence alignment'.

Reading a gene tree

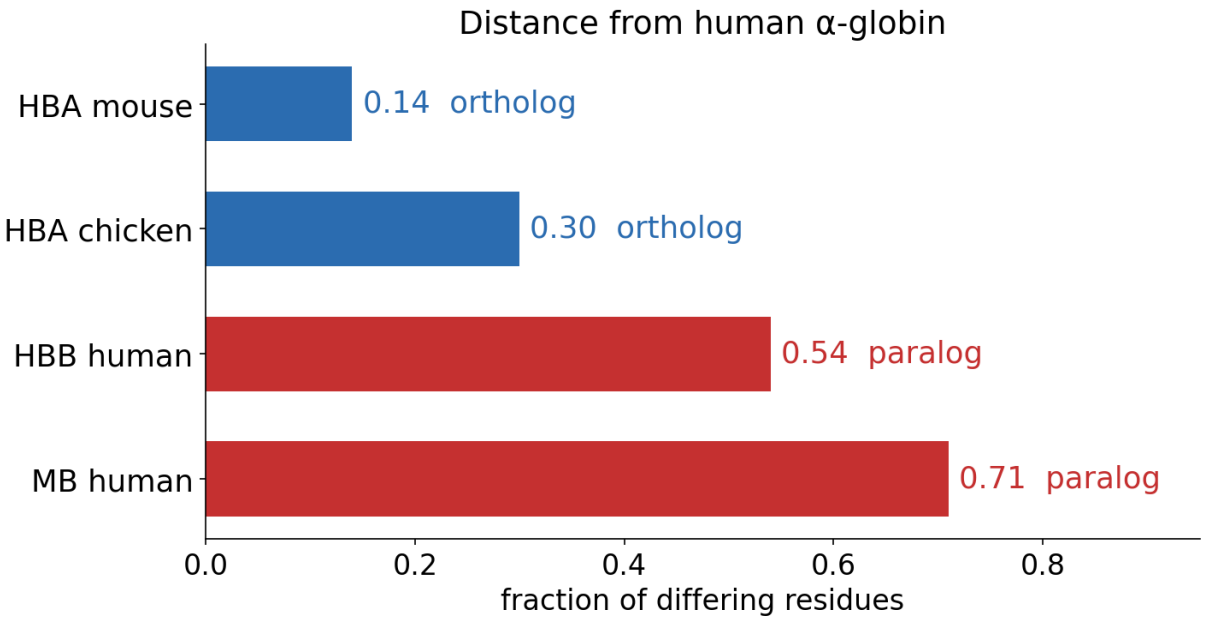


- **Leaves: sequences. Nodes: ancestors**
- Branch length: amount of change
- Midpoint root: order of events
- **Red: duplication. Blue: speciation**

Neighbour joining in the course notebook; see course.bioinfo.se/session04.html, 'From alignment to tree'.

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Orthologs are closer than paralogs

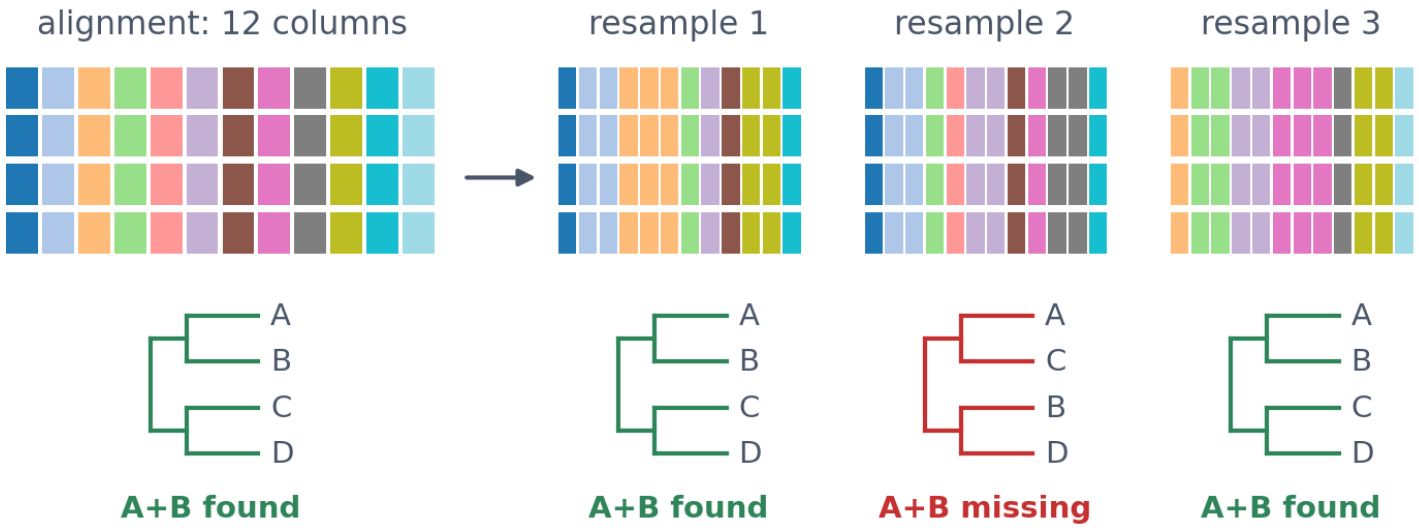


- **Human vs mouse α-globin: orthologs**
- **Human α vs β: paralogs**
- The tree, not the distance, decides

Distances printed by the course notebook.

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How much can you trust a branch? The bootstrap

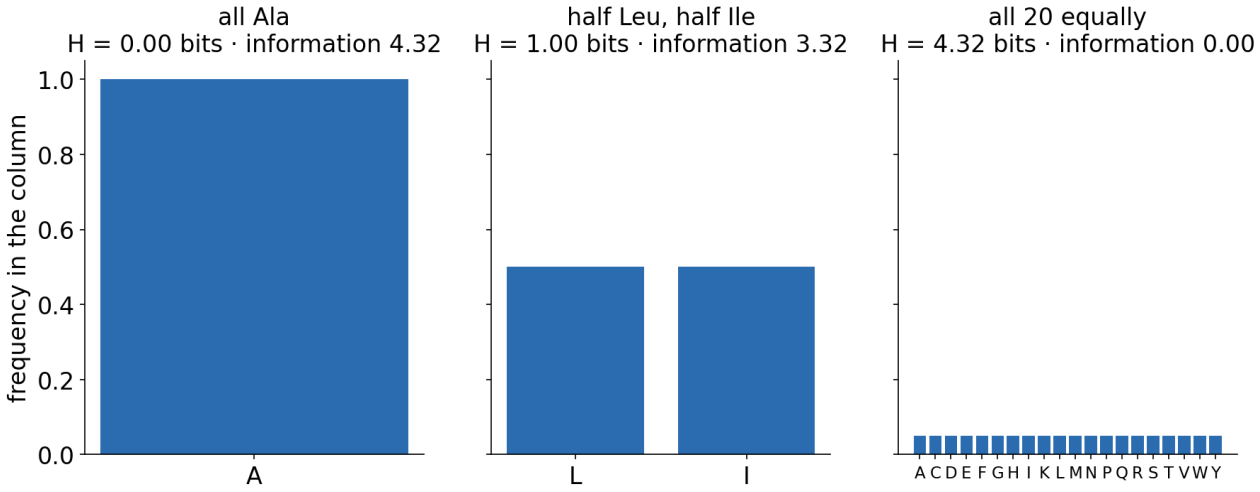


Schematic. Resample the columns with replacement, rebuild the tree each time: support for A+B = 2 of 3

Resample the alignment columns with replacement, rebuild the tree, and count how often each branch returns. Real phylogenetics adds substitution models and maximum likelihood (IQ-TREE, FastTree).

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Entropy: how variable is a column?

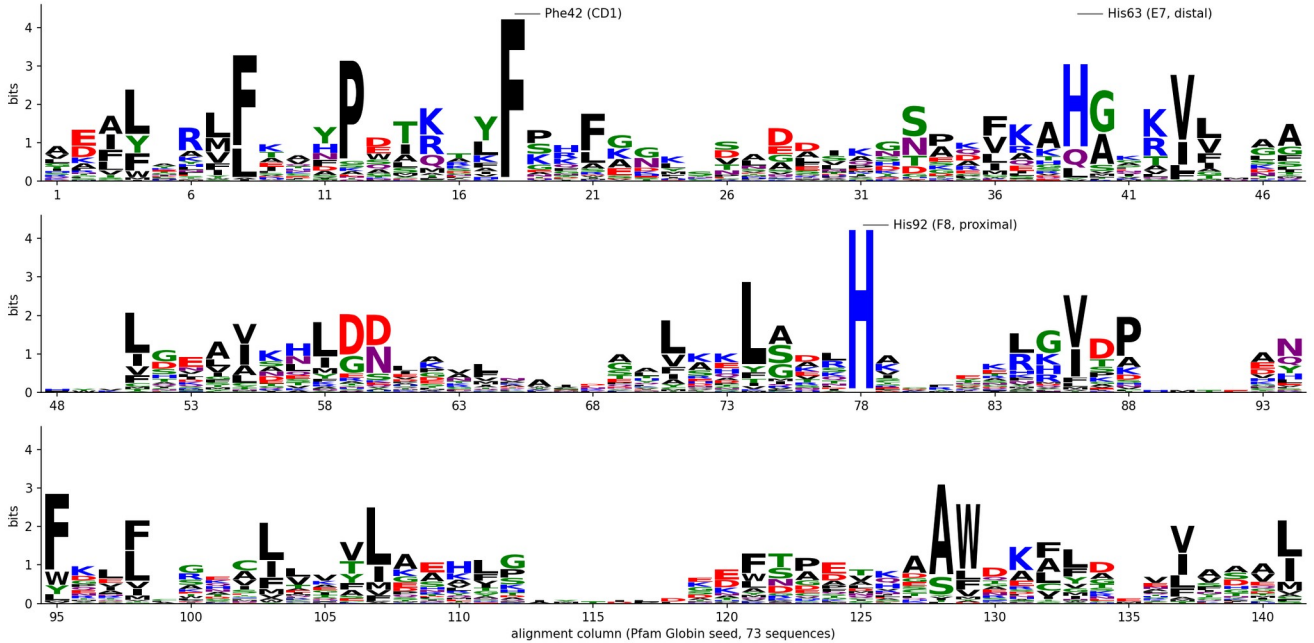


- **$H = -\Sigma\ p \log_2 p$**
- 0 bits: fully conserved
- 4.32 bits: all 20 equally common
- **Information = 4.32 − H**

See course.bioinfo.se/session04.html, 'Reading an alignment: entropy and sequence logos'.

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73 globins: what evolution keeps



Pfam Globin seed alignment. Stack height: information content; letter height: frequency. Phe42 and the proximal His92 (haem iron ligand) are identical in all 73; the distal His63 in 73%.

Drawn from Pfam PF00042 (InterPro) with logomaker; tools/builders/make_session04_logo.py.

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The fold is kept, the sequence is not

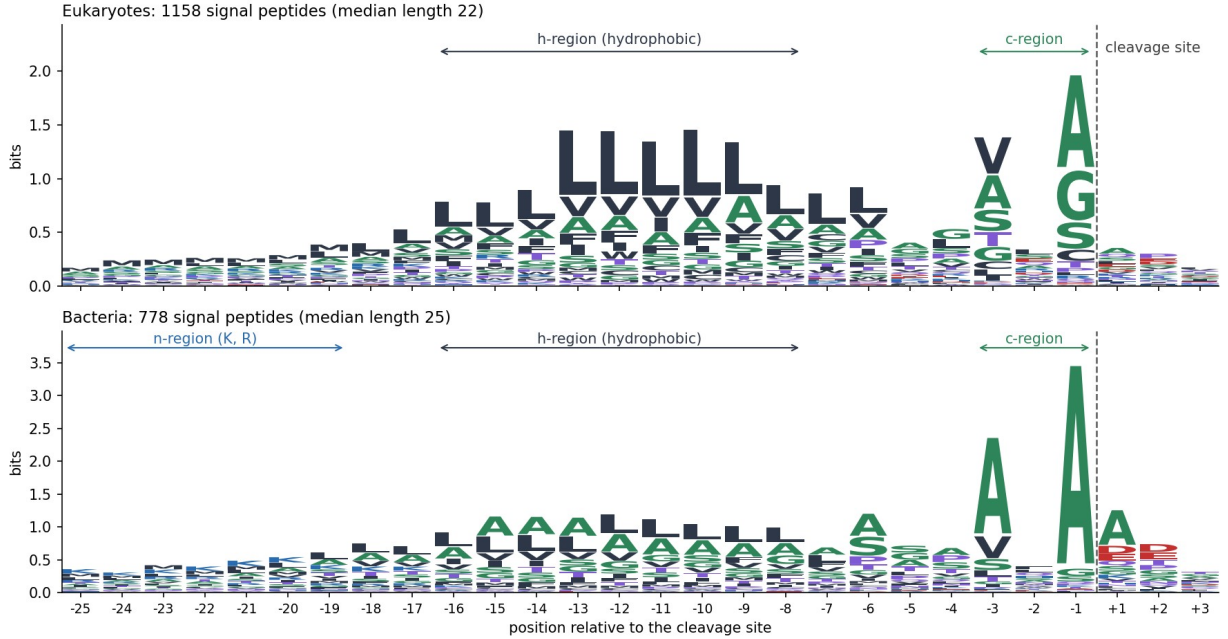
- **Only two columns identical in all 73**
- Hydrophobic columns (black) hold the fold
- Few sequences: columns look conserved by chance

1.2 bits
the average column

< 20%
identity, most pairs of globins

See course.bioinfo.se/session04.html, 'Reading an alignment: entropy and sequence logos'.

Signal peptides: a motif without a family

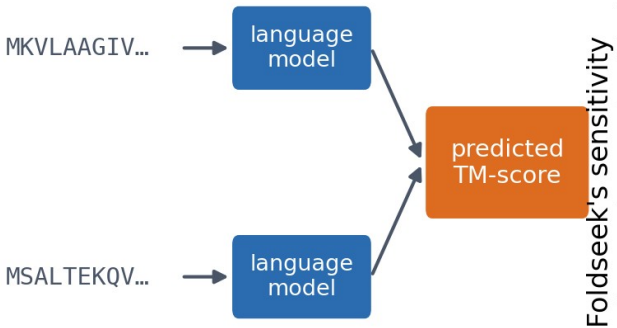


Unrelated proteins, aligned on the cleavage site instead of by homology. h-region: no column conserved, but all lean hydrophobic (L). Positions −3 and −1: small residues, the (−3, −1) rule. n-region (K, R): clearest in bacteria.

von Heijne (1983), Eur J Biochem 133:17; data: the Session 11 set (UniProt, experimental evidence).

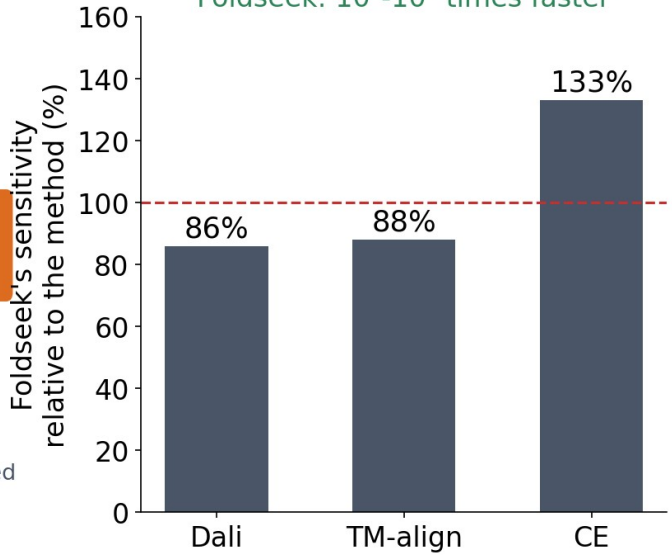
Alignment without a substitution matrix

Learned similarity from sequence



TM-Vec: no substitution matrix, no structure needed

Foldseek: 10^4-10^5 times faster

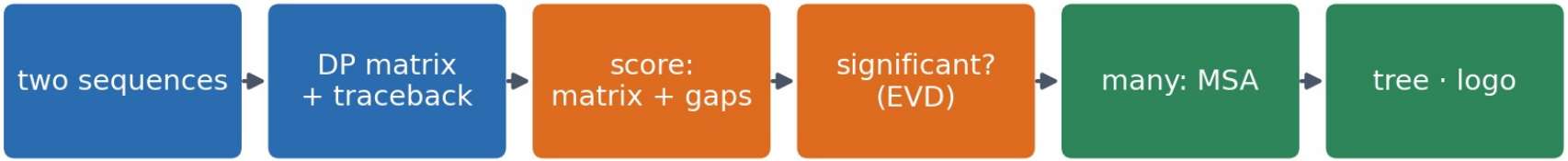


- Language models compare sequences
- TM-Vec: predicts structural similarity from sequence
- Foldseek: structures aligned 10^4-10^5 times faster
- **Is a learned similarity still a hypothesis about evolution?**

Hamamsy et al. (2023), Nature Biotechnology, doi:10.1038/s41587-023-01917-2; van Kempen et al. (2024), Nature Biotechnology 42:243.

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From two sequences to a family



Session 5: each MSA column becomes a score (a profile)

Dynamic programming finds the optimal alignment for a scoring scheme; the scheme (matrix and gaps) is a choice; a high score still has to beat chance. Many sequences give an MSA, a tree and a logo.

See course.bioinfo.se/session04.html, 'What's next'.

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What does the UPGMA guide tree do?

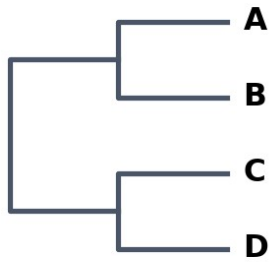
True history ((A, B), (C, D)), but B evolved fast. Progressive alignment builds its guide tree from the distances by UPGMA.

distances

	A	B	C	D
A	–	0.7	0.3	0.3
B	0.7	–	0.8	0.8
C	0.3	0.8	–	0.2
D	0.3	0.8	0.2	–

B evolved fast: far from everything

true history



((A, B), (C, D))

- **A** A and B first: most recent ancestor
- **B** C and D first, then A and B as relatives
- **C** C and D first, then A; B last
- **D** B first: it is the most different

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

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What does the UPGMA guide tree do?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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What does the UPGMA guide tree do?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session04-discussion-3.ipynb

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What does the UPGMA guide tree do?

Answer again

Same question, after the discussion.



quiz.bioinfo.se/live

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What does the UPGMA guide tree do?

Results, and who explains

Both rounds side by side.

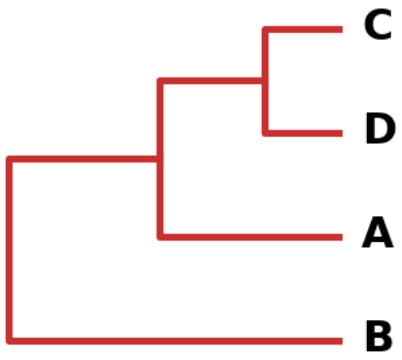
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 3 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session04-discussion-3.ipynb

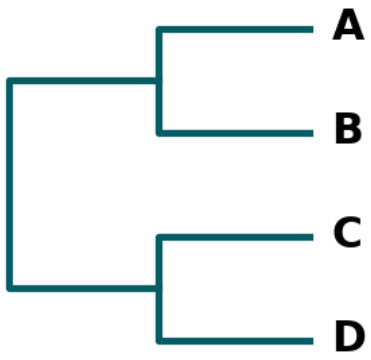
The answer: C and D, then A; B is added last

UPGMA guide tree



1. C + D (0.2) 2. A (0.3) 3. B last

true history (neighbour joining)



A and B are relatives

UPGMA assumes a molecular clock, so B, far from everything only because it evolved fast, is added last and A ends up with C and D. Neighbour joining recovers A and B as relatives. For ordering an alignment that is acceptable; as a phylogeny it is wrong.

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In up to three words: what was the hardest part of today?

Pairwise alignment

dynamic programming, global and local

Scoring

gaps, BLOSUM, significance

Many sequences

MSA, guide trees, logos

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

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The hardest part of today

What the room found hardest

Show the results ▸ quiz.bioinfo.se/live/report

Session 5 next

Needleman-Wunsch does not scale to searching a whole database
BLAST's heuristic is the answer, and its E-value builds on today's significance idea
Today's lab covers both Session 4 and Session 5